



Technical Requirements Document (TRD)

Project Name: Serene Web Platform **Focus:** Full-Stack Architecture, AI Integration, and Security

1. Technology Stack

- **Frontend Web:** React (Next.js) with Tailwind CSS for the "Organic Calm" UI.
- **Backend API:** Python FastAPI (highly performant, natively supports ML/AI libraries).
- **Database:** PostgreSQL (Supabase) with the `pgvector` extension.
- **ML Engine:** Scikit-Learn, XGBoost, SHAP.
- **AI Orchestration:** LangGraph, LangChain, `instructor`, and `pydantic`.
- **LLM Providers:** Hugging Face Inference API / Local Ollama (Phi-4/Llama-3).

2. Security Architecture (Mandatory)

Health data privacy is absolute. Antigravity agents **MUST** strictly adhere to:

1. **SQL Injection Prevention:** *Zero raw SQL*. All DB interactions use SQLAlchemy ORM or Supabase Python Client (parameterized queries).
2. **Rate Limiting:** Implement Redis-based rate limiting via `slowapi` (Auth: 5 req/min, Core: 60 req/min).
3. **Authentication:** JWT stored in secure, `HttpOnly`, `Secure`, `SameSite=Strict` cookies. Passwords hashed via Argon2/bcrypt. OAuth for sign up using google.
4. **CORS & Data Isolation:** Strict CORS matching the frontend domain. LLM payloads must never contain raw PII (Personally Identifiable Information).

3. AI Pipeline Architecture

- **Component 1 (ML):** Tabular user data is passed to XGBoost. Outputs `stress_score` and SHAP drivers.
- **Component 2 (LangGraph):** Planner and Analyzer nodes translate ML metrics into validated `AdvancedDashboardInsights` JSON.
- **Component 3 (Aura RAG):** Chat logs chunked and embedded via `mxbai-embed-large` into Supabase `pgvector`.
- **Component 4 (RLHF):** LLM-III generates nudges; implicit/explicit user responses adjust future prompt weights.